

Skills Summary

Separable Differential Equations with Initial Conditions

Use this reference while completing your worksheet or when studying.

Separable means $\frac{dy}{dx} = g(x)h(y)$ — the two variables can be pulled apart.

The three moves: (1) separate into $\frac{dy}{h(y)} = g(x) dx$; (2) integrate both sides, with a single $+C$; (3) apply the initial condition to turn C into a number.

General vs particular: a general solution still carries C ; a particular solution is the one curve through the given point. An initial-value problem is not finished until C is a number.

Useful antiderivative: $\int \frac{dy}{y} = \ln |y|$ — keep the absolute value until the initial condition tells you the sign.

1. Separate the variables and integrate

Everything with y (including dy) goes left; everything with x (including dx) goes right. If y cannot be isolated from x by multiplying and dividing, the equation is not separable and this method does not apply.

Example: Separate and integrate $\frac{dy}{dx} = \frac{x}{y}$.

Step 1. Both variables appear in a single quotient, so cross-multiplying splits them cleanly — separate before integrating anything.

Step 2. $y dy = x dx$. Integrating the left side with respect to y gives $\int y dy$, and the right gives $\frac{x^2}{2}$. Left side: $y^2/2$, so the implicit solution is $\frac{y^2}{2} = \frac{x^2}{2} + C$.

Try it: For $\frac{dy}{dx} = 5y$ the separated form is $\frac{dy}{y} = 5 dx$. What is the antiderivative of the left side?

check: $\ln(Abs(y))$ — keep the absolute value

2. Apply the initial condition to find C

Substitute the given pair into the equation you have *right after integrating* — before solving for y — and solve for C . Doing it early avoids carrying an unknown constant through a square root or an exponential, which is where sign errors appear.

Example: $\frac{dy}{dx} = 2x$ with $y(1) = 5$. Find C .

Step 1. The general solution is one integration away, so integrate first and let the condition do the only remaining work.

Step 2. $y = x^2 + C$. Substituting $x = 1$, $y = 5$ gives $5 = 1 + C$. $C = 4$, so the particular solution is $y = x^2 + 4$.

Try it: $\frac{dy}{dx} = 4x^3$ with $y(1) = 7$. What is C ?

check: $y = x^4 + C$ so $7 = 1 + C$ so $C = 6$

3. Use the particular solution

Once C is a number, the solution is an ordinary function: evaluate it, solve it for t , or read its long-run behaviour. For growth and decay, $\frac{dA}{dt} = kA$ always gives $A = A_0 e^{kt}$; a half-life h means $k = -\frac{\ln 2}{h}$. When solving for a time, divide out the initial amount *first* so only a ratio goes into the logarithm.

Example: The particular solution of a growth model is $y = 3e^{2x}$. Find $y(1.5)$.

Step 1. The constant is already pinned down, so this is now just function evaluation — no calculus left to do.

Step 2. $y(1.5) = 3e^{2(1.5)} = 3e^3$, and $e^3 \approx 20.0855$. $y(1.5) = \mathbf{60.26}$ to two decimal places.

Try it: A sample of 100 mg has a half-life of 5 days, so $A = 100e^{-(\ln 2/5)t}$. How much is left after 15 days?

check: $100e^{-(\ln 2/5)15} = 12.5$ mg (three half-lives)

(!) Watch out: a square-root solution has two branches and the initial condition picks one — if $y(1) = 2$ then $y = +\sqrt{x^3 + 3}$, never the negative root. And the interval of validity is the one containing the initial point: $y = \frac{1}{2-x}$ from $y(1) = 1$ is a solution only for $x < 2$.